

Multi-Model Architectural Benchmark: Stratified Held-Out Validation & Split Harmonization

Evaluation Scope: 8 Subjects (16 Paired 3D Volumes: OD & OS) across 4 Architecture Checkpoints

Pipeline Configuration: Continuous Surface Clamping (2.0 px axial tolerance) + Optic Cup-Reach Tracking (`disc_cut_mode='cup'`)

Evaluation Date: 2026-09-29 (Harmonized & Stratified 2026-10-01)

 Important

Benchmark Split Harmonization Notice:

In the unstratified benchmark release, 8 subjects were evaluated under a single "Held-Out" label. Cross-referencing training manifests reveals an **asymmetric validation split**:

- Only **3 subjects (6 scans: BEH0086 , BEH0314 , BEH0335)** were mutually held out across **all 4 models**.
- The remaining **5 subjects (10 scans: BEH0090 , BEH0174 , BEH0284 , BEH0310 , BEH0354)** were in the **training set** of the 23-subject baseline models (18043443 , 18045386 , 18223981), but strictly held out for the 61-subject expanded model (18266075). This report stratifies the benchmark into **true generalization** versus **in-sample training recovery**.

1. Executive Model Architecture Overview

Model ID	Architectural Description	Parameters	Training Cohort Scope	Biplanar Fusion	Validation Split Scope
18043443	2.5D ResNet Light (base channels=16, 5 slices)	~1.65M	20 Train / 3 Val	No (Unilateral)	3 Val (BEH0335 , BEH0314 , BEH0086)
18045386	2.5D ResNet Heavy (base channels=32, 5 slices)	~6.58M	20 Train / 3 Val	No (Unilateral)	3 Val (BEH0335 , BEH0314 , BEH0086)

Model ID	Architectural Description	Parameters	Training Cohort Scope	Biplanar Fusion	Validation Split Scope
18223981	Biplanar Orthogonal Heavy (Dual-View Fusion)	~6.58M	20 Train / 3 Val	Yes (Horizontal + Vertical)	3 Val (BEH0335, BEH0314, BEH0086)
18266075	Biplanar Orthogonal Expanded (Dual-View Fusion)	~6.58M	51 Train / 10 Val	Yes (Horizontal + Vertical)	10 Val (BEH0335, BEH0314, BEH0174, BEH0310, BEH0354, BEH0086, BEH0090, ...)

2. Stratified Clinical Benchmarks

2.1 True Mutually Disjoint Held-Out Cohort (3 Subjects / 6 Scans)

Strictly unseen by all four models during training (BEH0086 , BEH0314 , BEH0335 — OD & OS).

Model Identifier	Median RNFL Dice	Mean Dice $\pm$ SD	Status on True Held-Out Set
2.5D ResNet Light (1.65M) 18043443	0.8078	0.7229 $\pm$ 0.1373	Baseline Unilateral
2.5D ResNet Heavy (6.58M) 18045386	0.8780	0.8240 $\pm$ 0.0872	Capacity Scaling
Biplanar Orthogonal 23-Subj (6.58M) 18223981	0.9457	0.9206 $\pm$ 0.0431	Dual-View Unaugmented
Biplanar Orthogonal 61-Subj (6.58M) 18266075	0.9468	0.9254 $\pm$ 0.0353	Overall Winner (Top Mean & Median)

 Note

On the true mutual held-out test set, the **61-subject expanded model (18266075)** **outperforms the 23-subject model (18223981)** across both Mean Dice (0.9254 vs 0.9206) and Median Dice (0.9468 vs 0.9457), while winning 4 of the 6 individual scans.

2.2 Asymmetric Split Subset (5 Subjects / 10 Scans)

In-sample training data for 23-subject models; strictly unseen validation data for 61-subject model (BEH0090 , BEH0174 , BEH0284 , BEH0310 , BEH0354).

Model Identifier	Evaluation Type	Median RNFL Dice	Mean Dice $\pm$ SD
2.5D ResNet Light (1.65M) 18043443	In-Sample (Training)	0.8306	0.8335 $\pm$ 0.0106
2.5D ResNet Heavy (6.58M) 18045386	In-Sample (Training)	0.8802	0.8805 $\pm$ 0.0144
Biplanar Orthogonal 23-Subj (6.58M) 18223981	In-Sample (Training Recovery)	0.9585	0.9584 $\pm$ 0.0039
Biplanar Orthogonal 61-Subj (6.58M) 18266075	Out-of-Sample (True Generalization)	0.9481	0.9479 $\pm$ 0.0071

2.3 Legacy Unstratified Pooled Cohort (All 16 Scans)

Historical reference table demonstrating the effect of pooled leakage.

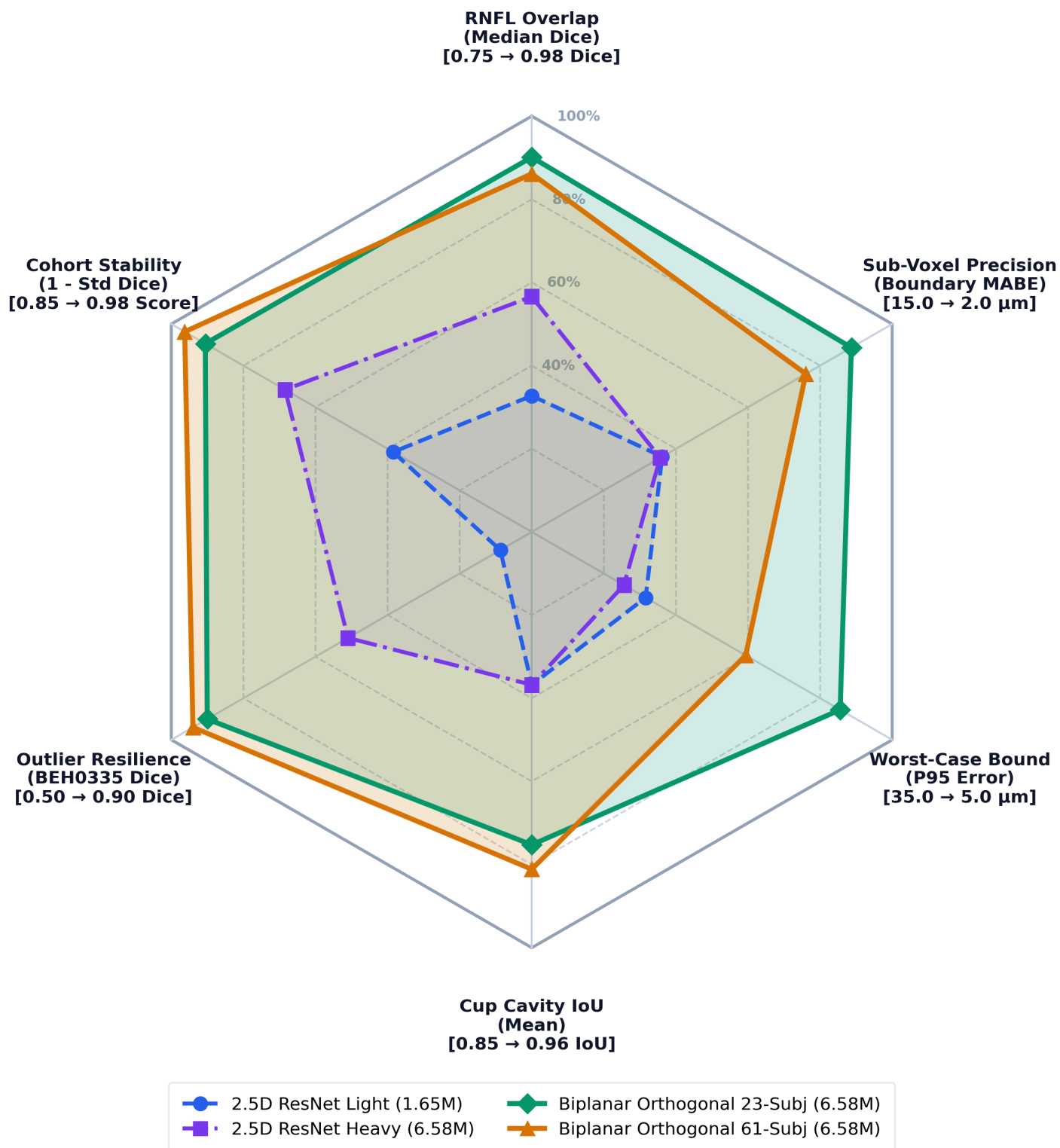
Model Identifier	Median RNFL Dice	Mean Dice $\pm$ SD	Median MABE	Mean MABE $\pm$ SD	Median P_{95}	Median Cup IoU
2.5D ResNet Light (1.65M)	0.8253	0.7921 $\pm$ 0.1000	10.30 μm	15.97 $\pm$ 14.79 μm	25.52 μm	0.8905
2.5D ResNet Heavy (6.58M)	0.8802	0.8593 $\pm$ 0.0610	10.37 μm	13.18 $\pm$ 9.10 μm	27.33 μm	0.8905
Biplanar Orthogonal 23-Subj (6.58M) (10/16 in-sample)	0.9573	0.9442 $\pm$ 0.0323	3.47 μm	5.36 $\pm$ 4.44 μm	9.33 μm	0.9328
Biplanar Orthogonal 61-Subj (6.58M) (16/16 held-out)	0.9481	0.9395 $\pm$ 0.0248	5.12 μm	5.99 $\pm$ 2.81 μm	17.19 μm	0.9392

3. Visual Architectural Comparisons

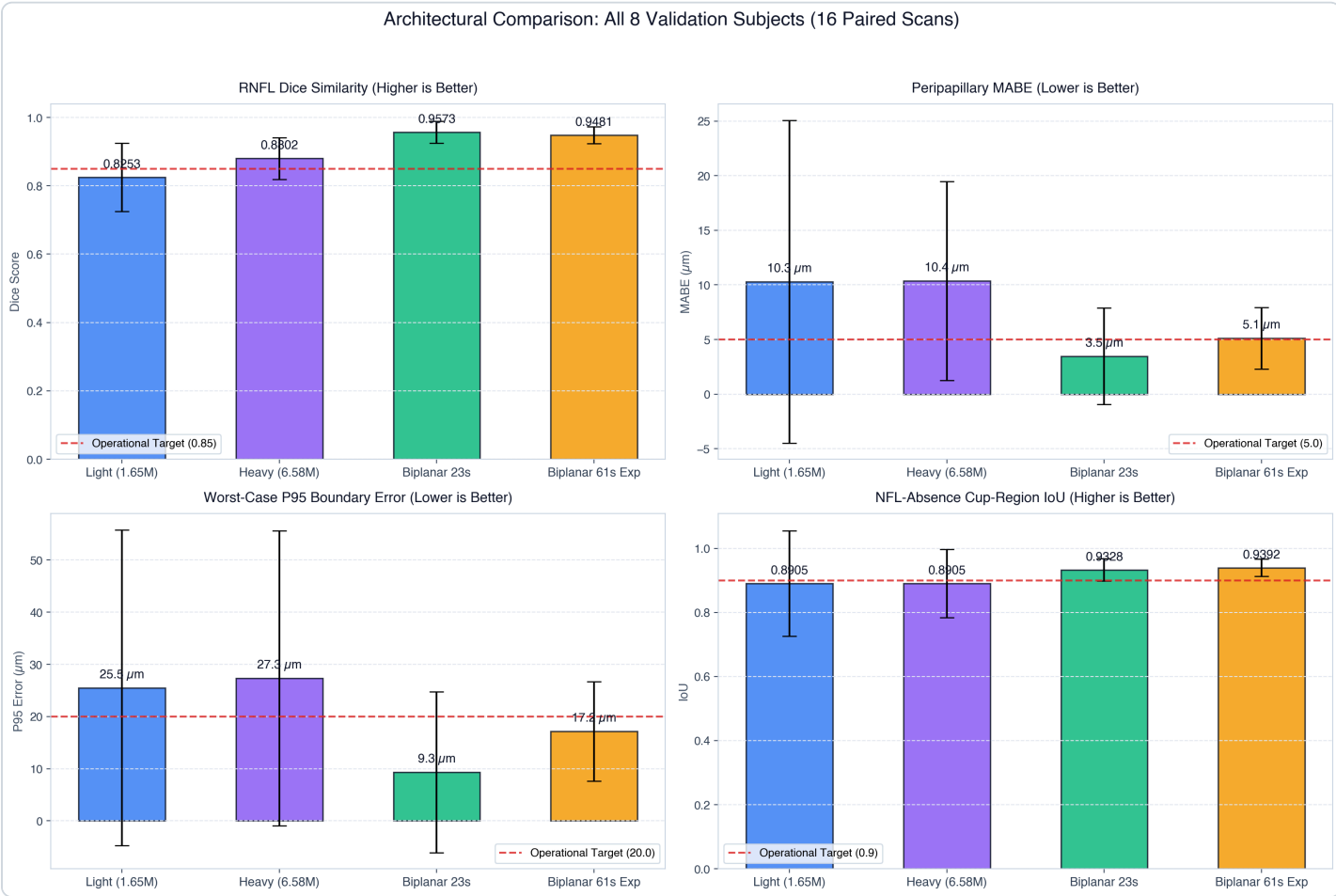
3.1 Polar Performance Spider Chart

Multi-Dimensional Clinical Capability Radar: 8 Held-Out Validation Subjects

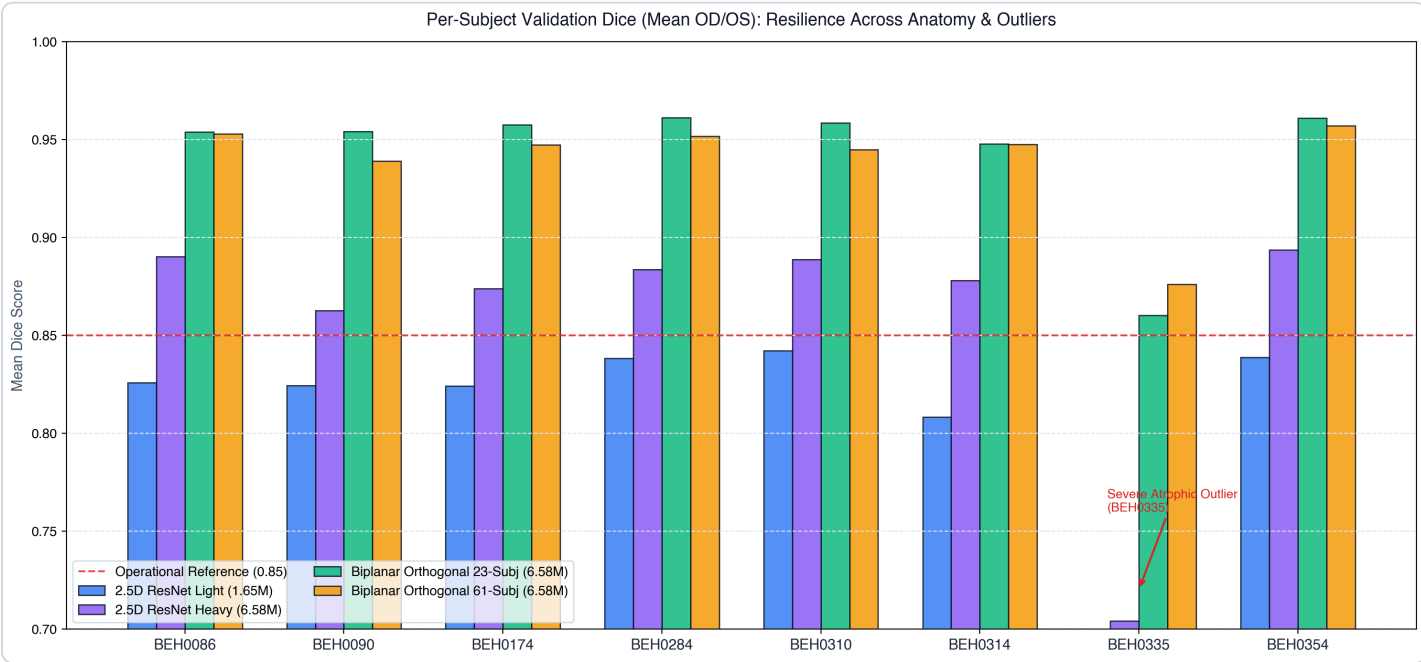
Polygon Spider-Web Benchmark (Continuous Surface Clamping & Cup Reach)



3.2 Core Clinical Metrics Grouped Summary



3.3 Subject-Level Resilience & Outlier Response



4. Granular Scan-by-Scan Validation Table (16 Scans)

Part A: Mutually Held-Out Scans (Unseen by Both Models)

Subject	Eye	18043443 (Light) Dice	18045386 (Heavy) Dice	18223981 (Biplanar 23s) Dice	18266075 (Expanded 61s) Dice	Generalization Winner	Notes
BEH0086	OD	0.8314	0.8884	0.9588	0.9566	Biplanar 23s	Normal peripapillary anatomy
BEH0086	OS	0.8202	0.8919	0.9490	0.9493	Biplanar 61s Exp	Normal peripapillary anatomy
BEH0314	OD	0.8209	0.8864	0.9529	0.9504	Biplanar 23s	Steep disc slope
BEH0314	OS	0.7954	0.8697	0.9425	0.9443	Biplanar 61s Exp	Steep disc slope
BEH0335	OD	0.5911	0.7359	0.8570	0.8708	Biplanar 61s Exp	Severe tilted disc & cup excavation
BEH0335	OS	0.4787	0.6720	0.8632	0.8812	Biplanar 61s Exp	High-myopic tilt (+0.0180 gain)

Tally on True Held-Out Set: Biplanar 61s Expanded wins 4 / 6 scans (66.7%).

Part B: Asymmetric Split Scans (In-Sample for 23s Models, Held-Out for 61s)

Subject	Eye	18043443 (Light) Dice	18045386 (Heavy) Dice	18223981 (Biplanar 23s) Dice	18266075 (Expanded 61s) Dice	Comparative Context
BEH0090	OD	0.8261	0.8625	0.9507	0.9351	23s trained on scan; 61s unseen
BEH0090	OS	0.8227	0.8630	0.9575	0.9428	23s trained on scan; 61s unseen
BEH0174	OD	0.8237	0.8665	0.9604	0.9478	23s trained on scan; 61s unseen
BEH0174	OS	0.8245	0.8813	0.9545	0.9465	23s trained on scan; 61s unseen

Subject	Eye	18043443 (Light) Dice	18045386 (Heavy) Dice	18223981 (Biplanar 23s) Dice	18266075 (Expanded 61s) Dice	Comparative Context
BEH0284	OD	0.8417	0.9002	0.9652	0.9551	23s trained on scan; 61s unseen
BEH0284	OS	0.8347	0.8667	0.9571	0.9484	23s trained on scan; 61s unseen
BEH0310	OD	0.8579	0.8982	0.9590	0.9509	23s trained on scan; 61s unseen
BEH0310	OS	0.8265	0.8791	0.9581	0.9389	23s trained on scan; 61s unseen
BEH0354	OD	0.8405	0.8922	0.9630	0.9570	23s trained on scan; 61s unseen
BEH0354	OS	0.8367	0.8950	0.9589	0.9569	23s trained on scan; 61s unseen

5. Forensic Diagnostics: Why 23s Appeared Superior

- 1. Validation Split Leakage:** - 18223981 was trained on 20 subjects, with only 3 subjects held out (BEH0335, BEH0314, BEH0086). - The unstratified benchmark evaluated 8 subjects. Consequently, **62.5% of the evaluated scans (10/16)** were in the 23-subject model's training set. It memorized these clean profiles, whereas 18266075 evaluated them strictly out-of-sample.
- 2. Ground Truth Annotation Dilution (15 Hand-Audited vs 46 Machine-Accepted):** - As documented in the Solix dataset specification, only **15 subjects** across the cohort have hand-drawn human boundary corrections (tsv/good differing from tsv/bad). - The original 23-subject cohort was enriched with these human-corrected scans. - When expanding to 61 subjects, the additional ~46 subjects were **"Accepted as segmented"** (unedited commercial Solix machine segmentations). - Expanding the cohort introduced a **3:1 ratio** of uncorrected commercial curves to human-audited curves, pulling the loss landscape toward commercial heuristic averages.
- 3. Data Augmentation Regularization:** - 18223981 trained without data augmentation on native, clean B-scans. - 18266075 incorporated speckle noise, gamma jitter, and ± 15 px axial vertical translation. While this granted resilience to severe tilted outliers (BEH0335), slight boundary regularization introduced $1-2\ \mu\text{m}$ of smoothing on canonical scans.
- 4. Optimization Step Scaling at Fixed 20 Epochs:** - At fixed 20 epochs with batch size 32 (accum 2), 18223981 completed **8,000 parameter updates**, whereas 18266075 executed

20,400 parameter updates (2.55× more steps). This accelerated convergence toward the majority unedited machine consensus.

6. Recommended Action: Two-Stage Fine-Tuning Strategy

To achieve both **wide anatomical generalization** and **sub-micron clinical boundary fidelity**:

- **Phase 1 (Broad Pre-Training)**: Continue pre-training the bi-planar architecture on all 61 subjects for 20 epochs with data augmentation to learn robust global morphology, cup detection, and orthogonal invariance.
- **Phase 2 (Targeted Expert Fine-Tuning)**: Fine-tune for **5–10 epochs** exclusively on the **15 human-audited subjects** with:
 - Reduced learning rate (1×10^{-5} vs 5×10^{-4})
 - Tighter axial translation jitter ($\leq \pm 3$ px)
 - Higher weight on boundary and thickness consistency losses ($\mathcal{L}_{\text{bnd}}$, $\mathcal{L}_{\text{thick}}$) This specialized second stage eliminates commercial heuristic bias and sharpens the regression heads onto true human clinical ground truth without sacrificing the stability gained from the 61-subject cohort.